

Taking Your Poster from Zero to Hero Using AI

Taufeq Mohammed Razakh

*Thomas Lord Department of Computer Science
University of Southern California*

Email: razakh@usc.edu

Ideas to support a three minute pitch by you

Registering for a Poster Competition

- **Conferences and Symposia often have awards for various categories** — most impressive visualization, novel application of quantum computing, impactful climate modelling, audience poll, *etc.*
- **ACM Student Research Competition (SRC):** Undergraduate and graduate students present their original research at well-known conferences before a panel of judges and attendees.
- **Conferences where Special Interest Groups (SIG) meet:** SIGACCESS, SIGAI, SIGAPP, SIGARCH, SIGBED, SIGCHI, SIGCOMM, SIGCSE, SIGDA, SIGDOC, SIGGRAPH, **SIGHPC**, SIGMETRICS, SIGMICRO, SIGMOBILE, SIGOPS, SIGPLAN, SIGSOFT and SIGSPATIAL



STUDENT
RESEARCH
COMPETITION



Association for Computing Machinery
Advancing Computing as a Science & Profession

Program Structure (Timeline) @ PASC26

- **The Platform for Advanced Scientific Computing (PASC) Conference**
Announcement: Building Trust in Science through HPC Co-Design
(Theme)
- **Abstract Review:** 2 page extended abstract comprehensively evaluated by award committee will until available slots are filled
- **Attend**
- **Flash Poster Session:** Maximum 30 seconds to engage the audience
- **Main Poster Session:** hang physical poster on poster boards. Discussion with judges and audience throughout
- **Presentation Round:** Semi-Finalist present before a panel of judges with a supporting power point presentation
- **Grand Finals:** In April

PASC



Association for
Computing Machinery



CSCS

30 second elevator pitch

- **Flash Poster Session:** Maximum 30 seconds to engage the audience
- **Recording:** Watch the presenters pitch here:

<https://pasc-conference.org/pasc26/session/?sess=sess129>

Final Result: 1st Place in the Graduate Category



Start with 2-Page Abstract

Mixed Precision Acceleration of Light Matter Dynamics: Enabling HPC Co-design for Quantum Material Discovery

Razakh Taufeq Mohammed
Thomas Lord Department of Computer Science
University of Southern California
razakh@usc.edu

1. Motivation and Challenges

The 2023 Nobel Prize in Physics was given to Agostini, Krausz, and L'Huillier for their discovery of non-linear interaction of high-intensity laser with matter to generate ultrashort attosecond (10^{-18} seconds) pulses. This breakthrough has opened new scientific and technological opportunities for controlling electronic excitation dynamics at their fundamental time scales. It holds promise for addressing the enormous power demands of ubiquitous artificial intelligence (AI) by enabling ultralow-power, ultrafast computing and sensing devices based on topological quantum materials [1]. Achieving this vision requires a predictive understanding of laser-induced electronic excitation dynamics at the attosecond scale. However, modelling laser-induced ultrashort timescales requires solving multiple field and particle equations of light, electrons, and atoms across spatiotemporal scales. Light-matter interaction is described using nonadiabatic quantum molecular dynamics (NAQMD), which couples quantum dynamics (QD) of electrons with molecular dynamics (MD) of atoms through complementary first-principles approaches.

On the other hand, high-performance computing (HPC) machines have undergone rapid diversification. Exascale systems such as Aurora, El Capitan, and Frontier are built around heterogeneous architectures dominated by graphics processing unit (GPU) accelerators. GPUs are increasingly relying on matrix engines for lower-precision computing to better serve memory-intensive large language models (LLMs) and enable performance at scale [2]. Given that double precision is widely adopted in scientific computing, rethinking the role of reduced-precision arithmetic and resolving the performance-accuracy trade-off it poses could be key to multiscale/multiphysics simulation in the post-exascale era.

2. Multiscale Simulation and Precision Aware Linear Algebra

To study quantum dynamics, such as attosecond physics, a metascable (design-once, scale on future architectures) paradigm is adopted to address the multiscale / multiphysics / heterogeneity / low-precision challenge [3]. By applying this paradigm to the multiscale light-matter dynamics simulation package, DC-MESH (divide-and-conquer Maxwell-Ehrenfest-surface hopping), has allowed for the end-to-end simulation of optically controlled topotonics extending from atomic to device scales in a hardware-optimal manner on an Exaflop machine. The algorithmic innovations of DC-MESH are (1) *Divide-conquer-recombine (DCR)* algorithms [4] which not only divides the problem into spatial but also physical subproblems of different computational

characteristics, and (2) *Metamodel-space algebra (MSA)* which allows the key data structures of subproblems to reside in respective hardware units.

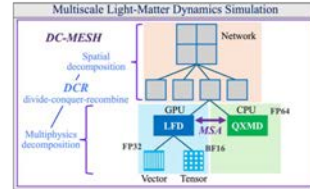


Figure 1: Divide-and-conquer domains for global potential in the DC-MESH method utilizing metamodel space algebra and mixed precision.

As illustrated in Figure 1, the divide-and-conquer domains are embedded in a global potential, with DC-MESH method consisting of (i) *Local Field Dynamics (LFD)* to describe light-electron interaction on GPU and (ii) *Quantum Excitation Molecular Dynamics (QXMD)* to describe electron-atom coupling on CPU, with minimal CPU-GPU data transfer via metamodel space algebra.

While chemical accuracy for the QXMD subproblem in DC-MESH requires FP64 precision, the shadow dynamics coupling from LFD has a limited dynamic range in the interval $[0,1]$ which can be achieved in FP32 without sacrificing accuracy. Furthermore, nonlocal correction in LFD is perturbatively added and is constructed to reproduce the dominant energy term exactly, which is amenable to even lower-precision GEMM (general matrix-multiply) calls. In fact, brain-floating-point 16 (BF16) with FP32 accumulation (hybrid FP32/BF16) is sufficient, as discussed in the performance vs. accuracy trade-off.

3. Accuracy and Performance Results

Accuracy Study Hardware: A dual-card node containing two Intel® Data Center GPU Max Series 1550. Each GPU has Intel® XMX enabled and contains 128 GB of High Bandwidth Memory (HBM). Each stack contains 448 EUs at up to 1.6 GHz frequency. The stack includes the Xe-core Tile (compute tile). Each compute tile comprises multiple Xe cores. Each Xe core has eight vector engines supporting FP64, FP32, and FP16 precisions, as well as Intel® XMX to support systolic numerics such as TF32, FP16, BF16, and INT8.

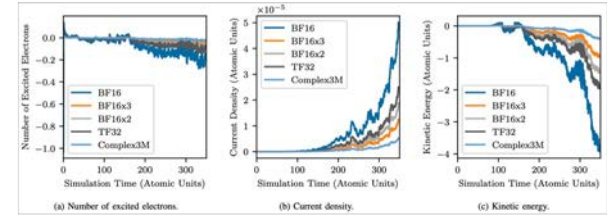


Figure 2: Accuracy of three output metrics, measured as the deviation from FP32 for BLAS calls using different compute modes

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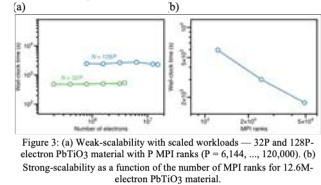
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Discussion: Studying three BF16 variants: BF16, BF16x2, and BF16x3, there is a trade-off between performance and accuracy as evident in Table 1 and Figure 2, with BF16 being the fastest and BF16x3 the most accurate. Given the system's high dynamical nature, as reflected in the rapidly increasing kinetic energy, it is acceptable. Furthermore, the energy of the entire 135-atom system is in the order of 10^3 Hartree. The deviation is seemingly much less striking for the case of the number of excited electrons, and negligible for the case of current density where deviation is measured in the order of 10^{-5} Atomic Units.

With the granularity of 128 electrons per MPI rank, the parallel efficiency is perfect 1.0 and within measurement fluctuation at $P = 120,000$ for 15,360,000-electron PbTiO₃ material in Figure 3 (a). This result demonstrates very high scalability of electronic solvers within the DCR algorithmic framework. The strong-scaling parallel efficiency is 0.843 with 98,304 MPI ranks for 12,582,912 electrons as shown in Figure 3 (b).

The 15,360,000-electron problem solved on 10,000 Aurora nodes sustains 1.873 EFLOP/s, which is 100.2% of FP64 peak or 70.1% of FP32 peak.



4. Conclusions

This work resolves an existing performance-accuracy dichotomy by proposing new algorithm-hardware co-design pathways for a wide variety of multiscale/multiphysics problems in the post-exascale era. For instance, complex polar topology in Figure 4 leverages insights from simulations to inform free-electron laser experiments for next-generation transducer sensor applications. This work was part of the 2025 ACM Gordon Bell Prize finalists.

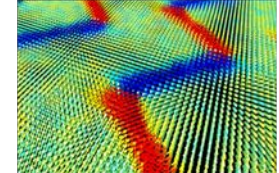


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5. REFERENCES

- [1] T. Linker *et al.*, Sci. Adv. 8, eabk2625 (2022).
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- [3] T. M. Razakh *et al.*, Proc. PDSEC (2024).
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Now make the 2-Page Abstract into a poster

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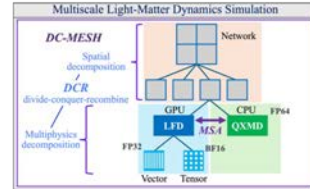


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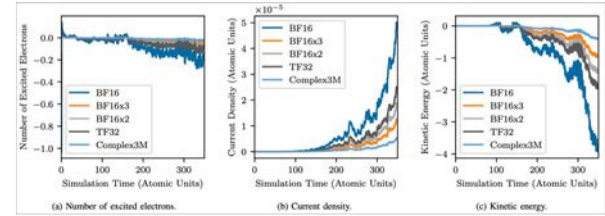


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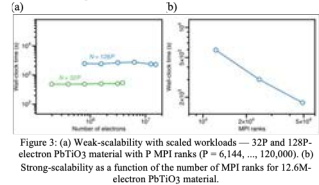


Figure 3: (a) Weak-scalability with scaled workloads — 32P and 128P-electron PbTiO₃ material with P MPI ranks (P = 6,144, ..., 120,000). (b) Strong-scalability as a function of the number of MPI ranks for 12.6M-electron PbTiO₃ material.

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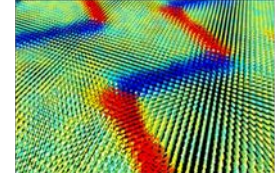


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Winning Poster ?

Mixed Precision Acceleration of Light Matter Dynamics: Enabling HP Co-design for Quantum Material Discovery

Tarek Mohamed Rashed
Thomas Lord Department of Computer Science



Motivation and Challenges

The 2023 Nobel Prize in Physics was given to Agazzi, Krauss, and Müller for their discovery of non-linear interaction of high-intensity laser with matter to generate ultrashort attosecond (10^{-18} second) pulses. This breakthrough has opened new scientific and technological opportunities for controlling electronic excitation dynamics at their fundamental time scales. It leads to new approaches for addressing the enormous power demands of algorithms and hardware (AI) for creating ultrashort pulses, simulating and analyzing devices based on topological quantum materials. Achieving this vision requires a profound understanding of laser-induced electronic excitation dynamics at the attosecond scale. However, modeling laser-induced ultrashort timescales requires solving multiple field and particle equations of light, electrons, and atoms across spatiotemporal scales. Light-matter interaction is described using time-dependent quantum molecular dynamics (TQMD), which couples quantum dynamics (QD) of atoms with molecular dynamics (MD) of atoms through various approximations. On the other hand, high-performance computing (HPC) machines have undergone rapid diversification. Ecosystems such as Aurora, El Capitan, and Frontier are built around heterogeneous architectures dominated by graphics processing unit (GPU) accelerators. GPUs are increasingly relying on matrix engines for lower-precision computing to better serve memory-intensive large language models (LLMs) and enable performance at scale. Given that double precision is widely adopted in scientific computing, reducing the role of reduced-precision arithmetic and meeting the performance-accuracy trade-off it poses could be key to multiscale/multiphysics simulation in the post-moore era.

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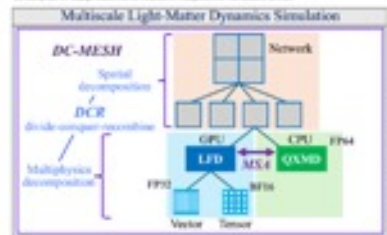


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As illustrated in Figure 1, the divide-and-conquer domains are embedded in a global potential, with DC-MESH method consisting of (i) **Global Field Dynamics** (GFD) to describe light-matter interaction on GPU and (ii) **Quantum electron Molecular Dynamics** (QEMD) to describe electron-atom coupling on CPU, with minimal CPU-GPU data-transfer via metamodel space algebra. While chemical accuracy for the QEMD subproblem in DC-MESH requires FP64 precision, the shadow dynamics coupling from GFD has a limited dynamic range in the interval $[0, 1]$ which can be achieved in FP32 without sacrificing accuracy. Furthermore, nonlocal correction in GFD is perturbatively added and is constructed to reproduce the dominant energy term exactly, which is amenable to even lower-precision GEMD general matrix-multiply calls. In fact, float-precision point 16 (BF16) with FP32 accumulation (hybrid FP32/BF16) is sufficient, as discussed in the performance in accuracy trade-off.

Accuracy Results

Accuracy Study Hardware: A dual-card mode containing two Intel® Data Center GPU Max Series T200. Each GPU has Intel® Xeon® enabled and contains 128 GB of high Bandwidth Memory (HBM). All experiments were run on a single stack to avoid Non-Uniform Memory Access (NUMA) effects. Each stack contains 448 EUs at up to 1.6 GHz frequency. The stack includes the 8-core Tile (compute EUs). Each compute EU comprises multiple fix cores. Each fix core has eight vector engines supporting FP64, FP32, and FP16 precision, as well as Intel® AVX to support scalar numerics such as TF32, FP16, BF16, and INT8.

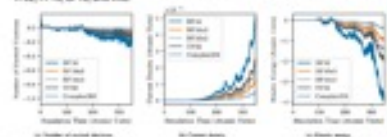


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Performance Results

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Discussion

Studying three BF16 variants, BF16, BF16x2, and BF16x3, there is a trade-off between performance and accuracy as evident in Table 1 and Figure 2, with BF16 being the fastest and BF16x3 the most accurate. Given the system's high dynamical nature, as reflected in the rapidly increasing kinetic energy, it is acceptable. Furthermore, the energy of the entire 128-atom system is in the order of 10^4 Atomic Units. The deviation is seemingly much less striking for the case of the number of excited electrons, and negligible for the case of current density where deviation is measured in the order of 10^{-10} Atomic Units. With the granularity of 128 electrons per MPI rank, the parallel efficiency is perfect 5.6 and within measurement fluctuation $\approx 1 = 120,000$ for 13,360,000 electron PzF103 material in Figure 3 (b). This result demonstrates very high scalability of electronic solvers within the DCR algorithm framework. The strong scaling parallel efficiency is 0.843 with 68,324 MPI ranks for 13,362,812 electrons as shown in Figure 3 (c). The 15,360,000-electron problem solved on 10,880 Aurora nodes sustains 1.471 GPU/GPU, which is 100.2% of FP64 peak or 70.1% of FP32 peak.

Figure 3: (a) Weak-scalability with scaled workloads — 53P and 108P-electron PzF103 material with P MPI ranks ($P = 5,144, \dots, 120,000$). (b) Strong-scalability as a function of the number of MPI ranks for 13.6M-electron PzF103 material.

Conclusion

This work results in a scaling performance-accuracy dichotomy for simulating new electronic materials design paradigm for a wide variety of multiscale/multiphysics problems in the post-moore era. For instance, complex error handling in Figure 4 originates from simulations to inform laser generation experiments for nanophotonics. This work was part of the 2023 ACM Gordon Bell Prize finalists.

Figure 4: Flux closure structure during ferroelectric switching in PzF103.



Prompt: How does this poster compare to the TUM Poster?



AI Response:

The TUM poster has:

- Tighter scientific storytelling;
- Cleaner hypothesis-driven narrative
- More focused experimental framing

Your poster has:

- Stronger HPC relevance
- Larger computational scale

Tighten your poster around one central question about precision-aware quantum dynamics, your work could become substantially stronger than the TUM example for ACM SRC.

Takeaway: (1) Needs better storytelling, and
(2) has a weak narrative arc!

You Need to Provide Context

Prompt (to give context about judging)

The theme for this year's conference is Building Trust in Science through HPC Co-Design

Response:

Exactly — and that changes the evaluation criteria substantially.

- Your current narrative is actually *better aligned* with the PASC 2025 theme than the TUM poster. The TUM poster is fundamentally about numerical precision analysis algorithmic stability, and mixed-precision experimentation



Takeaway: (1)Needs better storytelling, and
(2) ~~has a weak narrative are!~~

How much work can you offload to AI?

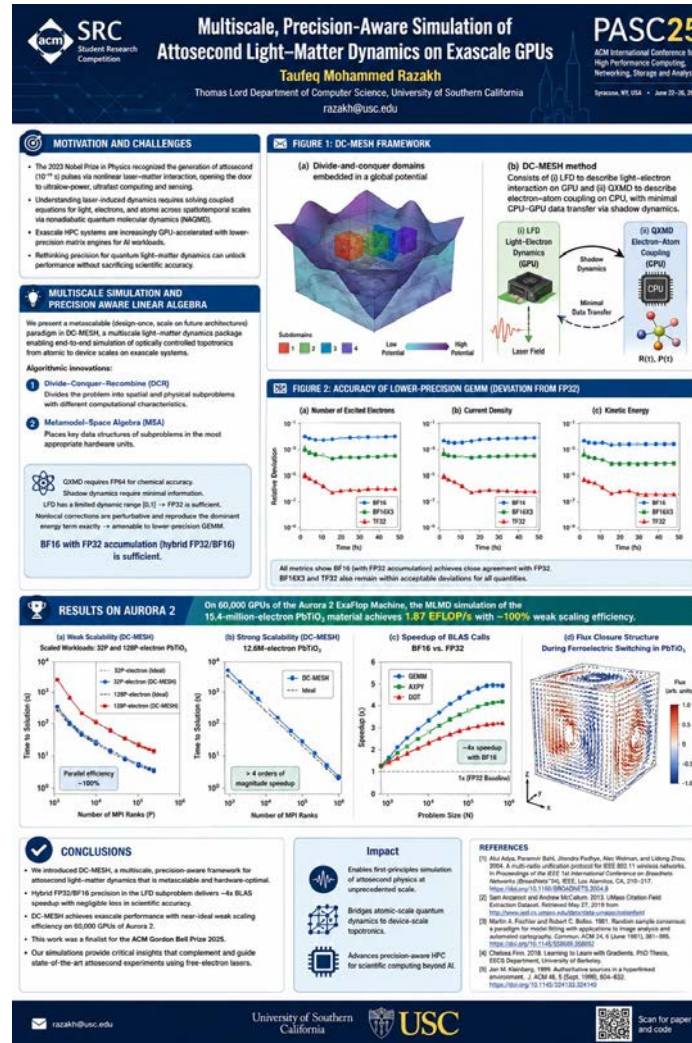
Prompt

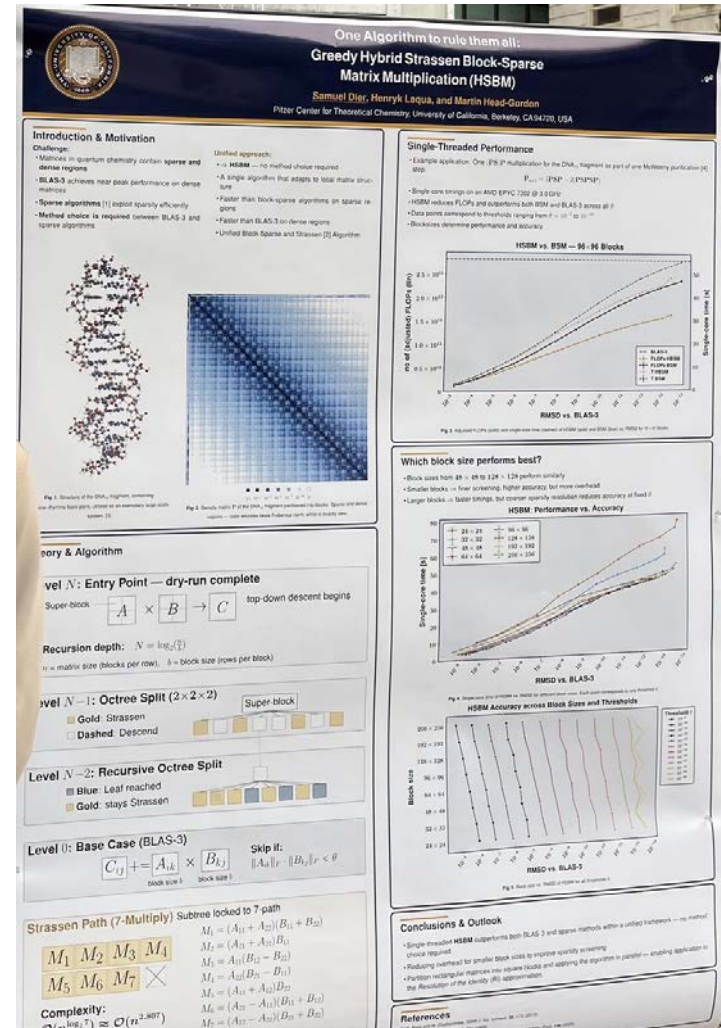
Now generate me the poster

Response:



Takeaway: (1) AI designs do not stand out; uniqueness is essential in competition.
(2) AI slop is evident





Provide more context into picture (contd.)

Prompt

Aim for about a

- A third text
- A third figures
- A third whitespace.
- consider people will be 6 feet (2m) away from the poster and want to get an overview in < 1

Constantin Brancusi
Bird in Space, 1927, cast later

On view: BCAM 3rd Floor Lobby

Negative space (Whitespace) in art:

In *Bird in Space*, the sculptor eliminated wings and feathers, elongated the swell of the body, and reduced the head and beak to a slanted oval plane



Use it like a Coach

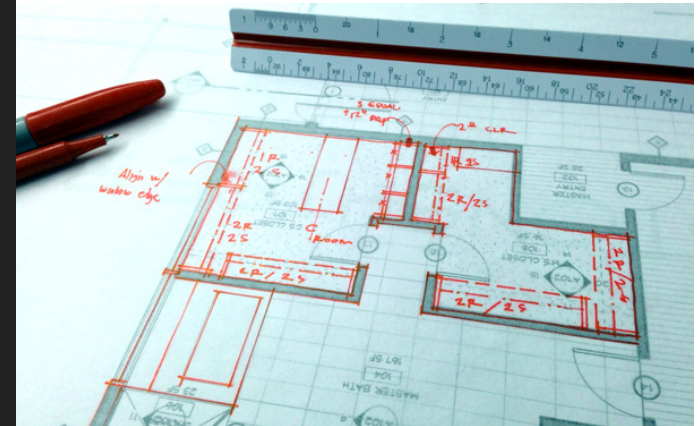
Prompt

Give me a redline/blueprint to use

Response:



Title		
One-sentence takeaway		
Motivation	Performance on Aurora	
DC-MESH Framework (Figure 1 - LARGE)	Scalability Results (Figure 3a,b)	
Accuracy Results (Figure 2)	Key Results	
	1.873 EFLOP/s	
	120,000 MPI ranks	
	3.9x BLAS speedup	
Figure 4 + Impact	Conclusions	
References	QR Code	Contact



Like a redline document useful during construction

Narrow scope to individual sections (Motivation)

Actively write excerpts for your sections and let AI provide feedback.

Prompt:

(Paste Paragraph)
and ask for bullet points

Response :

Motivation (Current: ~2 paragraphs → Reduce to 3 bullets)

Replace with:

- Why does this matter?
- Attosecond light–matter interactions enable ultrafast quantum devices.
- Simulating these processes requires coupled quantum and molecular dynamics.
- Existing simulations are too expensive for large-scale material discovery.



Be creative, explore adding elements

Prompt

Give me a callout box for MSA

Response:



My recommendation: Small callout attached to Figure 2

Put it immediately below or beside the DC-MESH diagram.

Something like:

What is MSA?

Metamodel Space Algebra
exchanges only essential
information between CPU and
GPU submodels, minimizing
communication while allowing
each component to execute at
its optimal precision.

This is enough.

Start seeing the bigger picture

Prompt

Show final (Motivation/Framework/Precision Aware) column

Response:



1. MOTIVATION & CHALLENGES

- Attosecond (10^{-18} s) light-matter interactions can enable ultra-low-power and ultra-fast computing devices.
- Simulating topological quantum materials requires solving coupled field and particle equations for light, electrons, and atoms.
- Exascale systems are increasingly GPU-driven and rely on lower-precision matrix engines.
- The performance-accuracy trade-off is a key challenge for multiscale and multiphysics simulations.

GOAL

Enable predictive quantum material discovery through algorithm-hardware co-design on exascale systems.

[Photo-switching / Skyrmion Figure]

Caption: Photo-induced switching in PbTiO_3 motivates predictive multiscale simulations.

2. DC-MESH FRAMEWORK

Physics Decomposition & Hardware Mapping

DC-MESH decomposes light-matter dynamics into independent physics subproblems and maps each component to the hardware where it performs best.

- DCR partitions the global simulation into independent subdomains.
- LFD models light-electron interactions on GPUs.
- QXMD models electron-atom coupling on CPUs.
- MSA minimizes communication between submodels.

[LARGE DC-MESH FIGURE]

KEY IDEA

Physics Decomposition + Hardware Mapping + Mixed Precision

- Reduced data movement
- Higher throughput
- Exascale scalability

3. PRECISION-AWARE EXECUTION

Balancing Accuracy & Performance

- QXMD requires FP64 precision for chemical accuracy.
- LFD operates efficiently in FP32.
- Perturbative nonlocal corrections are accelerated using BF16 tensor operations.
- Supported modes: FP64 / FP32 / {BF16, BF16x2, BF16x3}.

[Figure 3 Accuracy Plot]

The Final Product

Mixed Precision Acceleration of Light Matter Dynamics: Enabling HPC Co-design for Quantum Material Discovery

Taufeq Mohammed Razakh
Thomas Lord Department of Computer Science



Motivation and Challenges

- Attosecond (10^{-18} s) **light-matter interaction** can enable **ultra low power** and **ultra fast computing** devices
- **Simulating** topological quantum materials involves solving **multiple field and particle equations** for light, electrons, and atoms
- Exascale **systems** are increasingly graphics processing unit (GPU) driven, as such **relying on lower-precision** matrix engines
- **Performance-accuracy trade-offs** are key challenges facing multiscale and multiphysics simulations for the **post-exascale era**

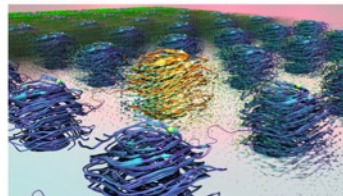
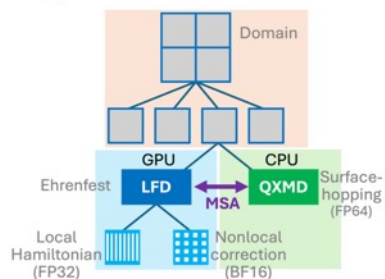


Photo-switching of PbTiO_3 motivates topological quantum material simulation

DC-MESH Framework

Divide-conquer-recombine (DCR) [1] algorithm **decomposes** light-matter **dynamics** into multiphysics sub-problems and maps it to hardware where it performs best

- **Quantum eXcitation Molecular Dynamics (QXMD)** models electron-atom coupling on **CPU**
- **Local Field Dynamics (LFD)** models light-electron coupling on **GPU** [2]



Divide-and-conquer domains for global potential in the DC-MESH method selectively using precisions

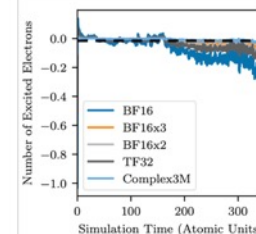
What is MSA? Metamodel Space Algebra links multiphysics models by exchanging essential information, while allowing each model to execute at optimal precision

Precision Aware Execution

- QXMD requires FP64 to preserve chemical accuracy during self-consistent field (SCF) iterations
- LFD operates in low dynamic range ($\Delta f_s^a \in [0,1]$) enabling FP32 execution
- Nonlocal corrections are accelerated using BF16 tensor operations
- Supported modes: **FP64 / FP32 / {BF16,BF16X2,BF16X3}**

Accuracy vs Performance

- The only quantity of interest is the **number of excited electrons**



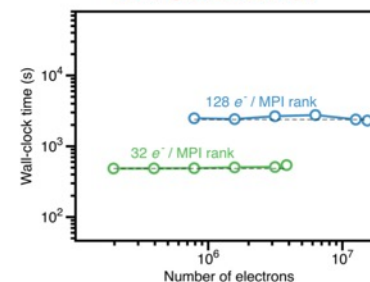
Accuracy of output metrics, measured as the deviation from FP32 for BLAS calls using different compute modes

- **BF16x3 provides best performance-accuracy trade-off**

Compute Modes	Measured Speedup	Theoretical Speedup
BF16	3.91x	16x
TF32	3.76x	8x
BF16x2	2.41x	(16/3)x
BF16x3	1.69x	(8/3)x
Complex_3M	1.12x	(4/3)x

Hardware? Aurora - 10,624 blades; Individual blade - 2x 2.00 GHz Intel®Xeon Max 9470 52-core CPUs, 64 GB HBM, 512 GB DDR5 Memory, 6x Intel®Data Center GPU Max 1550 GPUs.

Scaling into ExaFlop?



Weak scaling granularity of 128 electrons per MPI rank outperforms in mixed precision mode, perfect parallel efficiency on 120,000 ranks

- **1.87 EFlop/s on 60,000 GPUs;** atto-to-femtosecond laser-induced dynamics of **15.4-million electron system** [3]

Conclusion and Future Work

- **First multi-exaflop/s** performance sustained for **quantum-mechanical simulation**
- **MSA** unravels co-design pathways to **resolve performance-accuracy challenges** in the post-exascale era
- Future scope to **compare performance** gains when opting for **GEMM emulation** [4]

References

- [1] F. Shimojo et al., "A divide-conquer-recombine algorithmic paradigm for large spatiotemporal quantum molecular dynamics simulations," The Journal of Chemical Physics, 2014.
- [2] T. Linker et al., "Exploring far-from-equilibrium ultrafast polarization control in ferroelectric oxides with excited-state neural network quantum molecular dynamics," Science Advances, vol. 8, no. 12, p. eabk2625, 2022.
- [3] T. M. Razakh et al., "Multiscale light-matter dynamics in quantum materials: From electrons to topological superlattices," SC '25, 2025.
- [4] K. Ozaki et al., "Ozaki scheme II: A gemm-oriented emulation of floating-point matrix multiplication using an integer modular technique," arXiv: 2504.08009, 2025.

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